Amazon Sales Analytics Project Report

March – May 2022 | 80,000+ Orders

Power BI | SQL | Excel | DAX

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1. Project Overview

I built this project to analyze **Amazon India's sales data for 3 months — March, April and May 2022.** The dataset had around **79,353 order records** across different categories, states, and customer types.

The idea was simple — take real data, clean it properly, analyze it using **SQL and Power BI,** and find insights that can actually help the business make better decisions.

2. Why I Built This Project

Before jumping into the analysis, I wanted to be clear about what questions I was trying to answer:

- Which product categories are making the most money?
- Which states are our biggest markets?
- Why are so many orders getting cancelled and who is responsible — Amazon or Merchant?
- Are B2B customers any different from regular customers?
- What is causing product returns and can we reduce them?
- How did revenue change month by month across March, April and May?

3. DATA CLEANING & PREPARATION

Tool Used: Microsoft Excel

Before loading the data into Power BI, I cleaned it in Excel. Here is what I did:

- Removed duplicate records using **Excel Remove Duplicates function**
- **Filled blank values** in fulfilled-by column — Amazon fulfilled rows filled with **"Amazon"**, Merchant rows retained **"Easy Ship"**
- Cancelled orders had 0 in the Amount column — I kept these rows because they are important for cancellation analysis
- Fixed inconsistent values in Status, Category, and Size columns
- Created new column "Order Status" with 5 clean categories: **Shipped, Delivered, Cancelled, Returned, Pending**
- Created "Customer Type" column — B2B = TRUE mapped to "B2B", FALSE mapped to "B2C"

4. SQL Analysis

Tool: MySQL Workbench

After cleaning, I imported the data into MySQL and wrote 10 queries to validate my Power BI numbers and find insights.

Query 1 — Which category makes the most revenue?

```
SELECT Category,
ROUND(SUM(Amount), 2) AS Total_Revenue
FROM `amazon-sales`
GROUP BY Category
ORDER BY Total_Revenue DESC;
```

ANALYSIS- T-shirt came first with ₹24.52M, then Shirt ₹12.72M, then Blazer ₹6.52M.

Query 2 — How many orders are in each status?

```
SELECT `Order status`,
COUNT(DISTINCT `Order ID`) AS Total_Orders
FROM `amazon-sales`
GROUP BY `Order status`
ORDER BY Total_Orders DESC;
```

ANALYSIS- Shipped = 46,383 | Delivered = 20,216 | Cancelled = 11,229 | Returned = 1,517 | Pending = 8

Query 3 — Which states have the highest revenue?

```
SELECT `ship-state`,
ROUND(SUM(Amount), 2) AS Total_Revenue
FROM `amazon-sales`
GROUP BY `ship-state`
ORDER BY Total_Revenue DESC
LIMIT 10;
```

ANALYSIS- Maharashtra topped at ₹8.2M, Karnataka second at ₹6.3M.

Query 4 — Which fulfillment type cancels more?

```
SELECT Fulfilment,
COUNT(DISTINCT `Order ID`) AS Cancelled_Orders,
ROUND(COUNT(DISTINCT `Order ID`) * 100.0 /
(SELECT COUNT(DISTINCT `Order ID`) FROM `amazon-sales`), 2)
AS Cancellation_Rate_Percent
FROM `amazon-sales`
WHERE `Order status` = 'Cancelled'
```

GROUP BY Fulfilment;

ANALYSIS-- Amazon = 6,400 cancellations | Merchant = 4,300 cancellations. This was surprising because Amazon handles more orders but still cancels more.

Query 5 — Revenue split between Amazon and Merchant

```
SELECT Fulfilment,
COUNT(DISTINCT `Order ID`) AS Total_Orders,
ROUND(SUM(Amount), 2) AS Total_Revenue
FROM `amazon-sales`
GROUP BY Fulfilment
ORDER BY Total_Revenue DESC;
```

ANALYSIS- Amazon = ₹32M (67%) | Merchant = ₹16M (33%)

Query 6 — Which category gets returned the most?

```
SELECT Category,
COUNT(DISTINCT `Order ID`) AS Total_Returns
FROM `amazon-sales`
WHERE `Order status` = 'Returned'
GROUP BY Category
ORDER BY Total_Returns DESC;
```

ANALYSIS- T-shirt = 586 returns, Shirt = 534 returns, Blazer = 207 returns.

Query 7 — What is the overall return rate?

```
SELECT
ROUND(COUNT(DISTINCT CASE WHEN `Order status` = 'Returned'
THEN `Order ID` END) * 100.0 /
COUNT(DISTINCT `Order ID`), 2) AS Return_Rate_Percent
FROM `amazon-sales`;
```

ANALYSIS- Return rate = 1.92% — which is actually quite low and shows good product quality.

Query 8 — How different are B2B and B2C customers?

```
SELECT
CASE WHEN B2B = 'TRUE' THEN 'B2B' ELSE 'B2C' END AS Customer_Type,
COUNT(DISTINCT `Order ID`) AS Total_Orders,
ROUND(SUM(Amount), 2) AS Total_Revenue,
ROUND(AVG(Amount), 2) AS Avg_Order_Value
FROM `amazon-sales`
GROUP BY B2B;
```

ANALYSIS- B2B = 514 orders, average order value ₹749. B2C = 74,077 orders, average order value ₹645. B2B customers spend 16% more per order.

Query 9 — Expedited vs Standard shipping

```
SELECT `ship-service-level`,
COUNT(DISTINCT `Order ID`) AS Total_Orders,
ROUND(SUM(Amount), 2) AS Total_Revenue
FROM `amazon-sales`
GROUP BY `ship-service-level`
ORDER BY Total_Orders DESC;
```

ANALYSIS- Expedited = 49K orders | Standard = 25K orders. Most customers prefer faster delivery

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Query 10 — Monthly revenue trend

```
SELECT
MONTHNAME(STR_TO_DATE(Date, '%m-%d-%y')) AS Month,
MONTH(STR_TO_DATE(Date, '%m-%d-%y')) AS Month_Number,
ROUND(SUM(Amount), 2) AS Monthly_Revenue,
COUNT(DISTINCT `Order ID`) AS Total_Orders
FROM `amazon-sales`
WHERE Date IS NOT NULL AND Date != ''
AND MONTH(STR_TO_DATE(Date, '%m-%d-%y')) IN (3, 4, 5)
GROUP BY Month, Month_Number
ORDER BY Month_Number;
```

ANALYSIS- April peaked at ₹29M, May dropped to ₹20M — 31% decline.

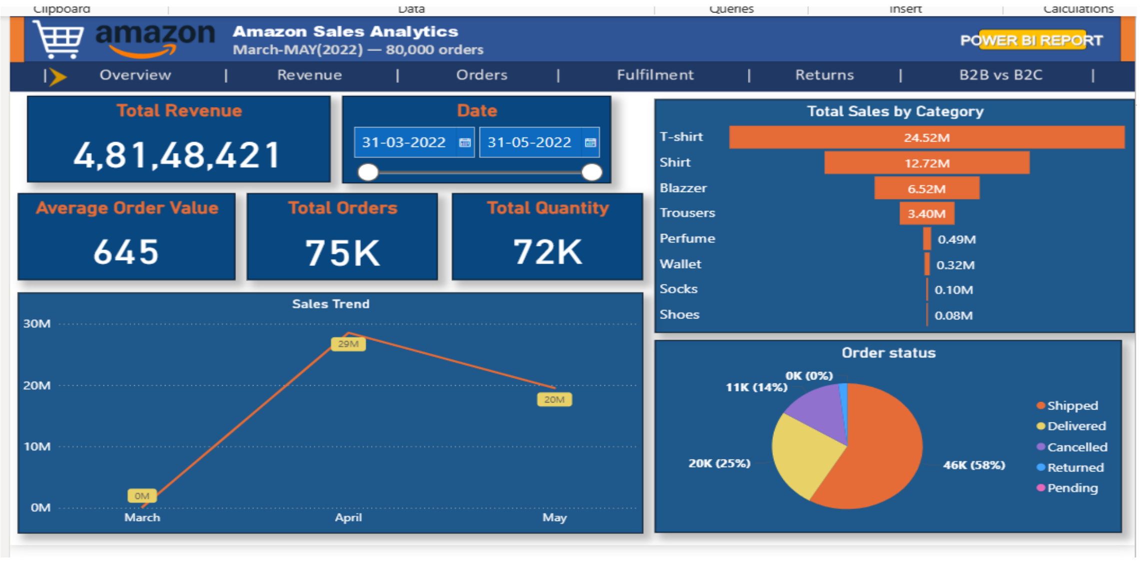
5. Key Numbers at a Glance

Total Revenue — ₹48.15M
Total Orders — 74,591
Total Quantity Sold — 72,000
Average Order Value — ₹645
Return Rate — 1.92%
Cancellation Rate — 14%
Amazon Fulfilled Orders — 50,113 (67%)
Merchant Fulfilled Orders — 24,478 (33%)
B2B Orders — 514
B2C Orders — 74,077
B2B Average Order Value — ₹749
B2C Average Order Value — ₹645

6. Dashboard Pages

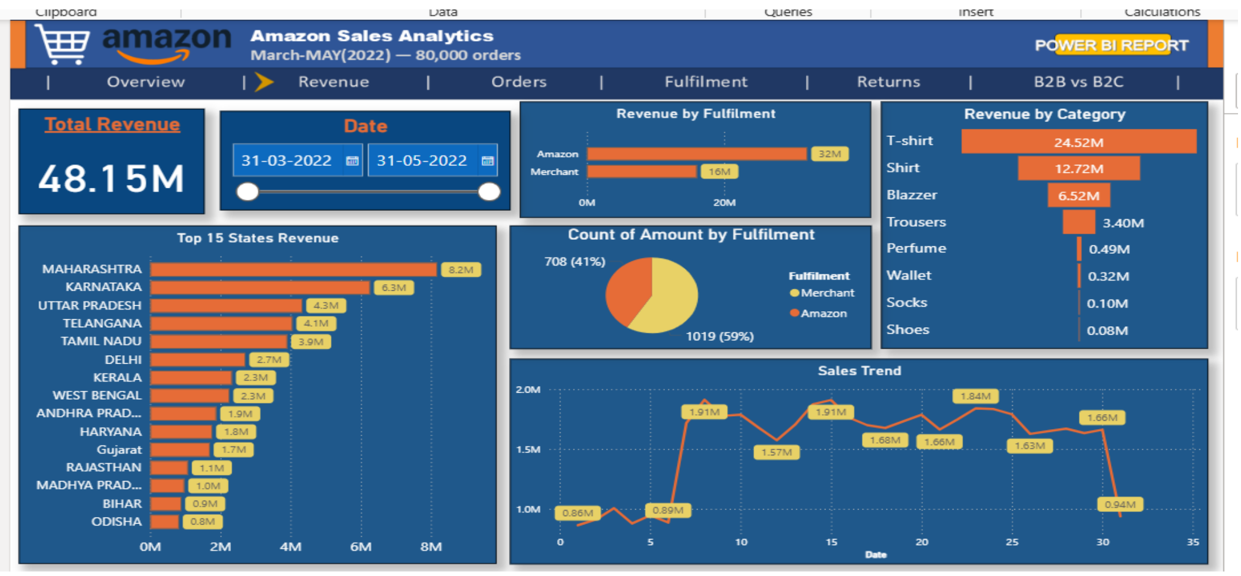
1. Overview

This page gives a quick summary of the entire business. I added KPI cards for revenue, orders, quantity, and average order value. The sales trend shows April was the best month. The order status donut shows most orders were shipped successfully.



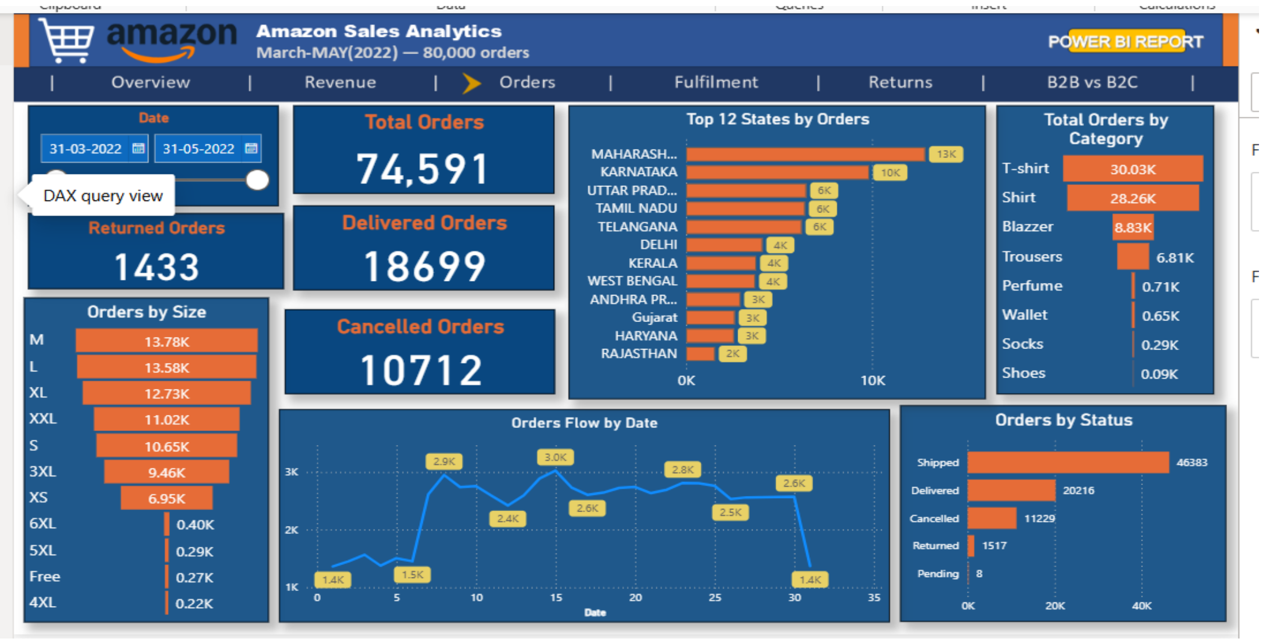
2 Revenue

This page breaks down where the money is coming from. T-shirt is clearly the top category. Maharashtra brings in the most revenue among all states. Amazon fulfilled orders contribute double compared to Merchant.



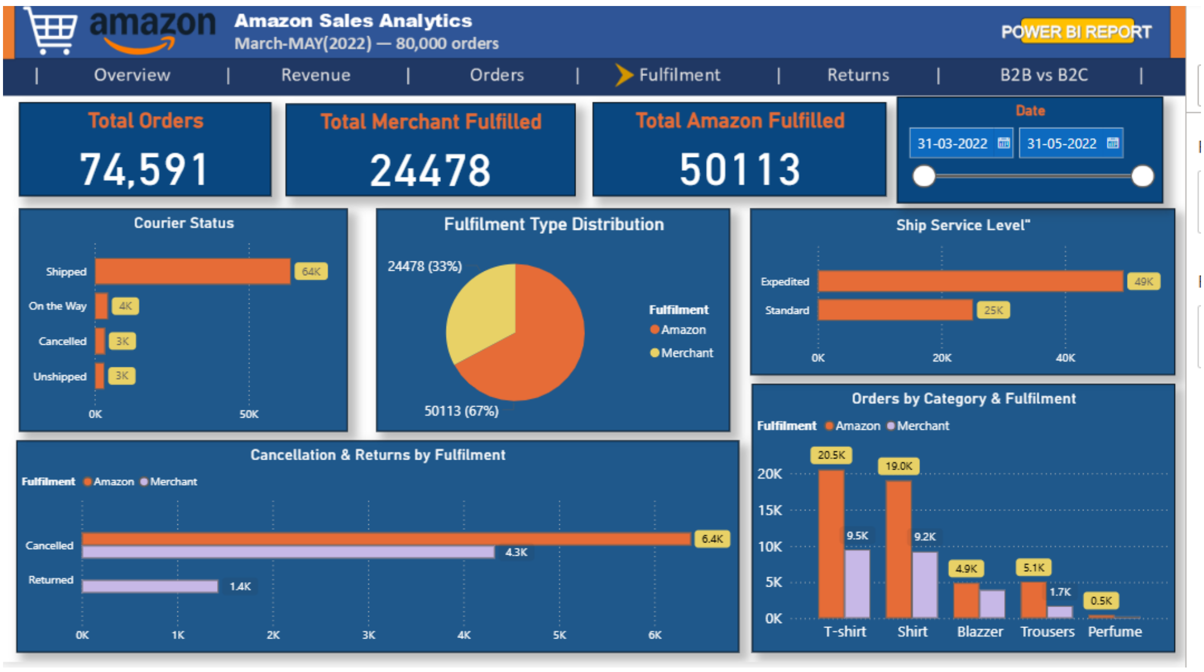
3 Orders

Here I looked at order volume by category, size, state, and status. Medium and Large sizes sell the most. T-shirt and Shirt dominate across all states.



4 Fulfillment

This was one of the most interesting pages. I found that Amazon cancels more orders than Merchant even though it handles more. But 100% of returns come from Merchant — Amazon has zero returns.

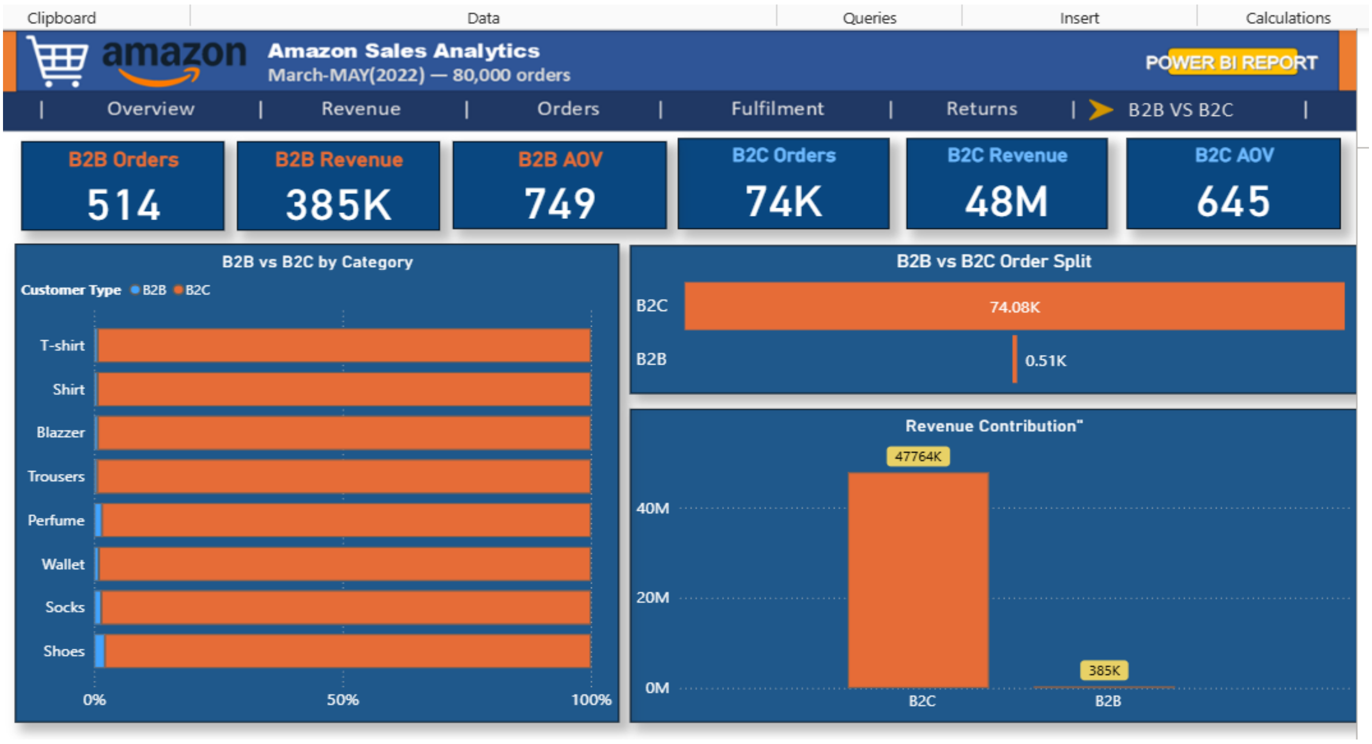


5 Returns

Only 1,433 orders were returned out of 79,353 — just 1.92%. T-shirt and Shirt are returned the most. All returns happened on Standard shipping — Expedited had zero returns.

6 B2B vs B2C

B2C is the main business — 74K orders and ₹48M revenue. But B2B customers are interesting — only 514 orders but they spend more per order (₹749 vs ₹645). This segment has potential to grow.



7. What I Found — Key Insights

Revenue:

→ T-shirt alone contributes 51% of total revenue

→ April was the peak month — revenue jumped to ₹29M then fell 31% in May

→ Maharashtra and Karnataka together contribute about 30% of total revenue

Fulfillment:

→Amazon cancels more orders than Merchant — 6,400 vs 4,300

→But Amazon has zero returns while Merchant accounts for 100% of returns

→This tells me Amazon is better at delivery quality but worse at preventing cancellations

Returns:

→Return rate is only 1.92% which is a positive sign

→T-shirt and Shirt together make up 78% of all returns

→Every single return happened on Standard shipping — not one return on Expedited

B2B vs B2C:

→B2B customers order less but spend 16% more per order

→This segment is currently only 0.69% of total orders — there is room to grow it

8. My Recommendations

1. Reduce Amazon Cancellations

Right now **Amazon 6,400 orders cancel** kar raha hai jo **Merchant ke 4,300** se zyada hai — even though Amazon double orders handle karta hai. This is **not a good sign**. A simple fix would be to track which drivers cancel the most and warn them. Also sending customers real-time order updates can help reduce last minute cancellations.

Next Step/Suggestion: Start tracking driver-wise cancellation rate every week.

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2. Offer Expedited Shipping More

This was the biggest finding in the whole project. Not a single order was returned on Expedited shipping. All **1,433 returns** came from Standard shipping only. Fast delivery clearly makes customers happier and reduces returns. If Expedited is made more

affordable or promoted more, **returns will naturally go down**.
Next Step/Suggestion: Run a 1 month trial offering discounted Expedited shipping and track if returns reduce.

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3. Put More Focus on Top 5 States

Maharashtra, Karnataka, Uttar Pradesh, Telangana and Tamil Nadu are giving the most orders and **revenue**. Instead of spending equally everywhere, putting more ads and offers in these states will give better results for the same budget.
Next Step/Suggestion: Increase marketing spend in these 5 states next quarter and compare results.

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4. Start Growing B2B Customers

B2B customers are only 514 right now but they **spend ₹749 per order** compared to **B2C's ₹645**. **B2B spend more** — we are just not focusing on them enough. Adding things like bulk pricing, GST invoices and a dedicated support number for business buyers can easily bring more B2B orders.

Next Step/Suggestion: Create a separate B2B landing page with bulk pricing options.

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5. Fix Size Issue in T-shirt and Shirt

These two categories alone make up **78% of all returns**. Most of the time customers return because the **size was wrong**. **Adding a proper size chart with actual measurements in centimeters, and encouraging customers to share size photos in reviews,** can help people **pick the right size and return less**.

Next Step/Suggestion: Update size charts for T-shirt and Shirt listings this month.

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6. Find Out Why May Sales Dropped

April revenue was ₹29M but May dropped to ₹20M — that is a **31% fall** in just one month. This could be **seasonal, or maybe a sale or promotion ended in April.** Understanding the reason is important so we can plan better next year and not be caught off guard.

Next Step/Suggestion: Compare April and May marketing activities and check if any offers or campaigns ended.

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9. DAX Measures I Used in Power BI

Total Orders — Counts unique order IDs using DISTINCTCOUNT on Order ID column

Total Revenue — Adds up all values in the Amount column using SUM

Avg Order Value — Divides Total Revenue by Total Orders to get average spend per order

Return Rate % — Divides returned orders by total orders and multiplies by 100

Amazon Fulfilled — Counts orders where Fulfilment column equals Amazon

Merchant Fulfilled — Counts orders where Fulfilment column equals Merchant

Cancelled Orders — Counts orders where Order Status equals Cancelled

Returned Orders — Counts orders where Order Status equals Returned

Delivered Orders — Counts orders where Order Status equals Delivered

B2B Orders — Counts orders where B2B column is TRUE

B2C Orders — Counts orders where B2B column is FALSE

B2B Revenue — Adds up Amount where B2B column is TRUE

B2C Revenue — Adds up Amount where B2B column is FALSE

B2B AOV — Calculates average order value for B2B customers only

B2C AOV — Calculates average order value for B2C customers only

Top Return Category — Finds the category with highest number of returned orders

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10. Conclusion

This was a good learning project for me. I got to work with a real-world dataset, clean it properly, run SQL queries to validate my numbers, build a 6-page Power BI dashboard, and write DAX measures for custom calculations.

The biggest thing I learned is that data can tell you things that are not obvious at first. I would not have guessed that Amazon — which handles more orders — actually cancels more than Merchant. Or that every single return happened on Standard shipping and not one on Expedited. These are the kinds of findings that make data analysis valuable.

If I had to pick three things I would tell the business team from this project it would be — fix the Amazon cancellation process, push Expedited shipping harder, and start treating B2B customers as a separate priority segment.